

Lakshya Bajaj

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Education

Purdue University, *West Lafayette, IN*

August 2024 – May 2028

B.S. in Robotics Engineering Technology; Concentration: Autonomy and UXVs (Dean's List x 3)

GPA: 3.78/4.00

Minors: Electrical Engineering Technology, Business Economics; Certificate: Collaborative Leadership

Relevant Coursework: Industrial Controls, Analog Signal Processing, Industrial Robot Programming, Graphic Communication & Spatial Analysis, Design Thinking in Technology

Projects and Research

Undergraduate Researcher, Mechanisms and Robotic Systems Lab, *West Lafayette, IN*

January 2026 – Present

- Designed schematics and custom PCB in KiCad for a 16×16 piezoresistive tactile array (256 nodes), implementing multiplexed readout and high-speed serial streaming (2 Mbps) for structured force data acquisition
- Developed signal conditioning pipeline including median baseline calibration, per-node thresholding, dynamic range clipping, fixed-scale normalization, and temporal low-pass filtering to improve measurement stability and reduce cross-talk
- Architected multi-threaded real-time visualization system (PyQtGraph + OpenGL) with cubic interpolation (16×16 → 64×64), Gaussian spatial smoothing, and 60 FPS surface rendering
- Extending tactile modeling framework to incorporate shear force estimation alongside normal forces for enhanced 3D force reconstruction in downstream learning-based inference pipelines

Robotics Engineer, Humanoid Robot Club, *West Lafayette, IN*

August 2024 – May 2025

- Contributed to the award-winning “chicken leg” foot architecture, earning the AMD Most Innovative Design of the Year award for mechanical–electrical integration and stability-driven design innovation
- Researched and designed electrical subsystems for humanoid legs, integrating motor drivers, IMUs, and microcontrollers for closed-loop control, real-time state estimation, and hardware validation during dynamic locomotion testing

Work Experience

Production Engineering Intern, NuTaste Food and Drink Labs, *Gurugram, India*

June 2025 – August 2025

- Designed and implemented a shift log sheet to track batch status, reducing handover time and improving process transparency; oversaw staff training on usage
- Applied lean manufacturing principles, including Kaizen and 5S methodology, to improve operational efficiency by 4%
- Collected and analyzed cycle time and process time data for labeling and capping, key bottlenecks in production
- Conducted a complete Value Stream Mapping exercise for one batch to identify inefficiencies and streamline operations
- Performed a cost-benefit analysis comparing square and round bottles, evaluating packaging density, truck space utilization, and equipment compatibility

Leadership Positions

Team Leader International, Purdue Orientation Programs, *West Lafayette, IN*

March 2025 – September 2025

- Mentored and guided 35 first-year students during Purdue's nationally recognized freshman orientation program
- Facilitated icebreakers, discussions, and team-building activities to ease cultural adjustment and build community
- Oversaw integration of international and domestic students, fostering cross-cultural connections and peer support
- Completed 30+ hours of leadership and cultural sensitivity training; contributed 90+ hours during orientation week

Skills

Robotics, Control & Automation: Industrial Robot Programming (FANUC KAREL, Yamaha Robot Language), PLCs, Kinematics, Motor Control, ROS, OpenCV, MediaPipe

Electronics & Hardware: Circuit Design, PCB Design and Fabrication (KiCad), Signal Conditioning, Power Systems

Embedded Systems & Sensing: Arduino, Raspberry Pi, Microcontrollers, Sensor Integration

Programming: Python, C++, C, MATLAB

Mechanical Design & Prototyping: Fusion 360, Siemens NX & Teamcenter, 3D Printing, Laser Cutting, Hand and Power Tools

Manufacturing & Process Optimization: Lean Manufacturing (Kaizen, 5S), Value Stream Mapping, Cycle Time Analysis, Cost-Benefit Evaluation